HOW MUCH DOES BODY POSE HELP PEDESTRIAN TRAJECTORY FORECASTING? A SEED-REPLICATED STUDY AND AUXILIARY-LOSS AUDIT ON WAYMO

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ABSTRACT

Body pose is a plausible but rarely used signal for pedestrian trajectory forecasting in autonomous driving. We extend the Motion Transformer (MTR) with a GRU encoder over past SMPL body poses (6D rotation representation) and measure what pose conditioning is actually worth on a pedestrian subset of the Waymo Open Motion Dataset, using LiDAR-derived pose pseudo-ground-truth. Our study has two parts. First, an *auxiliary-loss audit*: an earlier draft of this work credited geodesic SO(3) supervision with a 7.6% minADE improvement. We show analytically and empirically that in our data condition the geodesic loss receives *identically zero gradient*—future pose targets are absent—so the credited configurations differed only by random seed; the auxiliary signal that actually trains the encoder is a winner-takes-all L1 pose-regression loss. Second, a *seed-replicated measurement* of the surviving questions, with paired bootstrap confidence intervals over 6,688 validation pedestrians. Whether pose helps depends on the encoder, not the auxiliary loss: with the original GRU encoder, pose-conditioned training is heavy-tailed—8 replicates of one configuration span minADE 0.629–0.897—and on average *worse* than the no-pose baseline; with cross-attention over pose frames plus sinusoidal positional encoding, pose gives a stable -3.5% (95% CI $[-4.0, -3.0]$). Removing positional encoding removes the gain, confirming that temporal ordering carries the signal. HD map context reduces minADE by 26% on its own and shrinks the pose gain to -0.5% (p=0.046); after full-Waymo backbone pretraining (a further -22%), pose contributes -1.9% (CI $[-2.8, -1.0]$). Our results give a calibrated estimate of pose's value in a state-of-the-art forecaster, and a cautionary, reproducible example of how single-run ablations can attribute gains to inert components.

1 INTRODUCTION

Predicting where pedestrians will walk is a safety-critical problem in autonomous driving. State-of-the-art trajectory forecasting models such as MTR (Shi et al., 2022; 2024) achieve impressive accuracy by operating on observed agent positions, velocities, and HD map geometry—but they discard a potentially rich source of information: the *body pose* of the pedestrian.

Human gait encodes intent. A pedestrian who is turning to face a crosswalk will likely cross it; one whose torso is oriented away from traffic is unlikely to step into the road. A runner's body orientation predicts turns before the feet complete them. Modern perception systems increasingly provide body pose estimates (Zhu et al., 2023). While recent work uses skeleton keypoints for pose-conditioned prediction in robot navigation and pedestrian benchmarks (Salzmann et al., 2023; Saadatnejad et al., 2024; Gao et al., 2025), state-of-the-art autonomous driving forecasters operating on Waymo (Ettinger et al., 2021) discard body pose entirely. This paper asks: *does conditioning on past SMPL body poses improve pedestrian trajectory prediction within a SOTA AV forecasting framework, and if so, what design choices drive the gain?*

We extend MTR with a *pose conditioning branch* that encodes the past 11 timesteps of SMPL body pose (Loper et al., 2015) using 6D rotation representations (Zhou et al., 2019), a GRU encoder, and auxiliary future-pose prediction heads. We train and evaluate on a pedestrian subset of the Waymo Open Motion Dataset (Ettinger et al., 2021) paired with LiDAR-derived SMPL pose pseudo-ground-truth from Waymo-3DSkelMo (Zhu et al., 2025), using minADE and minFDE over 6 predicted modes.

This paper deliberately reports both a *result* and a *correction of our own earlier result*, because the correction is instructive. An earlier single-run draft credited geodesic SO(3) supervision with -7.6% minADE; an audit shows that was impossible—in our data condition the future pose targets are zero, so the geodesic loss is a constant with identically zero gradient and the two runs compared were the same experiment under different seeds (Section 4.3). We rebuilt the empirical claims with three to five seeds per configuration and paired bootstrap CIs over 6,688 validation pedestrians. Our corrected findings:

- **The encoder, not the auxiliary loss, decides whether pose helps.** Cross-attention over pose frames with sinusoidal positional encoding improves minADE by 3.5% (95% CI $[3.0, 4.0]$) with near-zero seed variance (0.6371 ± 0.0026). The GRU encoder used by the earlier draft is heavy-tailed: 8 replicates of its configuration span 0.629–0.897 (pooled mean 0.7060), *worse* than the baseline mean of 0.6604 ± 0.0143—a failure mode invisible at one seed per configuration.
- **Temporal ordering carries the pose signal.** The same cross-attention encoder *without* positional encoding is slightly worse than the baseline ($+1.0\%$): a bag of pose frames is useless. Auxiliary-loss composition, by contrast, is not the driver: no active auxiliary configuration reliably beats the no-auxiliary control (which itself edges out the baseline by 0.9%), and supervising with *real* future poses (motion-prior-completed, 99% coverage) does not help either.
- **Context dominates pose.** HD map features alone reduce minADE by 26% and full-Waymo backbone pretraining (231k pedestrian examples) a further 22% independent of pose. The incremental pose gains they leave (0.5% with map, 1.9% pretrained) are positive at the point estimates but not seed-level significant at three seeds (hierarchical bootstrap p=0.62, p=0.10).
- **Small-benchmark ablations need seed replication.** The spread among replicates of a single configuration (0.27 minADE across 8 seeds' extremes) exceeds every architectural effect except map and pretraining; one-seed comparisons on this benchmark can support nearly any narrative. We describe an audit procedure (loss-term gradient checks, target-coverage statistics, seed replication, per-pedestrian paired and hierarchical bootstrap) so it can be applied to similar pipelines.

2 Related Work

2.1 Trajectory Forecasting for Autonomous Driving

Modern trajectory prediction methods model multi-agent interactions over HD map features using transformer architectures (Vaswani et al., 2017; Shi et al., 2022; 2024; Yuan et al., 2021; Salzmann et al., 2020). Social GAN (Gupta et al., 2018) introduced generative multimodal prediction for pedestrians; MTR (Shi et al., 2022) advances this with global intention localization and local refinement via cross-attention over map and agent context. None of these methods incorporate body pose information.

2.2 Human Pose Estimation and Pose Representations

SMPL (Loper et al., 2015) provides a parametric body model where each joint rotation is represented as a 3D rotation matrix. For neural network learning, Zhou et al. (2019) demonstrate that 6D rotation representations (the first two columns of $SO(3)$) are continuous and better conditioned than quaternions or axis-angle, particularly under gradient-based optimization. The AMASS dataset (Mahmood et al., 2019) provides a large-scale archive of motion-captured body animations in SMPL parameterization and is widely used to train human motion priors (Rempe et al., 2021; He et al., 2022). Waymo-3DSkelMo (Zhu et al., 2025) combines LiDAR-based human mesh recovery (Fan et al., 2023) with such priors to produce temporally coherent SMPL pose sequences for Waymo pedestrians;

we use its output as pose pseudo-ground-truth. Works such as MotionBERT (Zhu et al., 2023) and HumanMAC (Chen et al., 2023) focus on pose quality but do not connect body pose to downstream trajectory prediction. A separate line uses 2D pose for pedestrian *crossing-intention* classification (Fang & López, 2018; Rasouli et al., 2019), supporting the premise that body configuration carries intent, though without producing metric trajectory forecasts.

2.3 Joint Pose and Trajectory Models

Several recent works condition trajectory prediction on body pose. Salzmann et al. (2023) leverage 3D skeletal keypoints from RGB-D cameras to improve trajectory prediction in robot navigation scenarios. Saadatnejad et al. (2024) integrate 2D/3D skeleton pose cues into a transformer-based predictor on pedestrian benchmarks, and Gao et al. (2025) extend this line to social navigation datasets using skeleton-based body language features. In contrast, our work uses parametric SMPL body pose in 6D rotation space—a richer, anatomically-grounded representation—within the state-of-the-art MTR framework on Waymo, and, with seed-replicated experiments and per-pedestrian confidence intervals, isolates which design choices actually drive the gain: auxiliary pose-loss design, temporal encoding (GRU vs. cross-attention $\pm$ positional encoding), future-pose supervision quality, and the interaction of pose with HD map context and backbone pretraining.

3 Method

3.1 Background: Motion Transformer (MTR)

MTR (Shi et al., 2022) is a transformer-based trajectory predictor for the Waymo Open Motion Dataset. For each *center object* (the agent being predicted), it encodes a scene context tensor from agent histories and HD map polylines via a PointNet-like encoder, then decodes $K = 6$ trajectory modes through iterative cross-attention over the scene context, guided by K learned intention queries that are globally localized to trajectory clusters.

The loss has three components: a mode classification loss $\mathcal{L}_{\text{cls}}$, a winner-takes-all regression loss $\mathcal{L}_{\text{reg}}$, and a velocity consistency loss $\mathcal{L}_{\text{vel}}$.

3.2 Pose Conditioning Branch

We add a pose conditioning branch alongside the trajectory decoder, illustrated in Figure 1. The branch has four components:

6D Rotation Encoding. Each SMPL body pose at timestep t consists of 24 joint rotations. We represent each rotation as a 6D vector (Zhou et al., 2019) (the first two columns of the 3×3 rotation matrix), giving a 144D pose vector per timestep.

GRU Pose Encoder. Given the past $T = 11$ pose vectors $\mathbf{p}_1, \dots, \mathbf{p}_T \in \mathbb{R}^{144}$, a two-layer GRU with hidden size 256 processes the sequence:

$$\mathbf{h}_t = \text{GRU}(\mathbf{p}_t, \mathbf{h}_{t-1}), \quad \mathbf{h}_T \in \mathbb{R}^{256}. \tag{1}$$

The final hidden state $\mathbf{h}_T$ is concatenated with the MTR agent feature and projected to 256D via an MLP, forming the fused agent representation.

Pose Prediction Heads. Each of the decoder layers additionally predicts K modes of future poses: at layer ℓ and mode k, a linear head outputs 6D rotations for all 80 future timesteps (a 80×144 output). These heads are auxiliary—their predictions never feed back into the trajectory pathway; their only effect on trajectory prediction is through the gradients their losses send into the shared decoder features and the GRU encoder.

Auxiliary Pose Losses. Let $\hat{\mathbf{r}}^{ij} \in \mathbb{R}^6$ be the predicted 6D rotation and $\mathbf{r}^{ij} \in \mathbb{R}^6$ be the ground-truth 6D rotation for joint j at future step i. We consider four auxiliary signals:

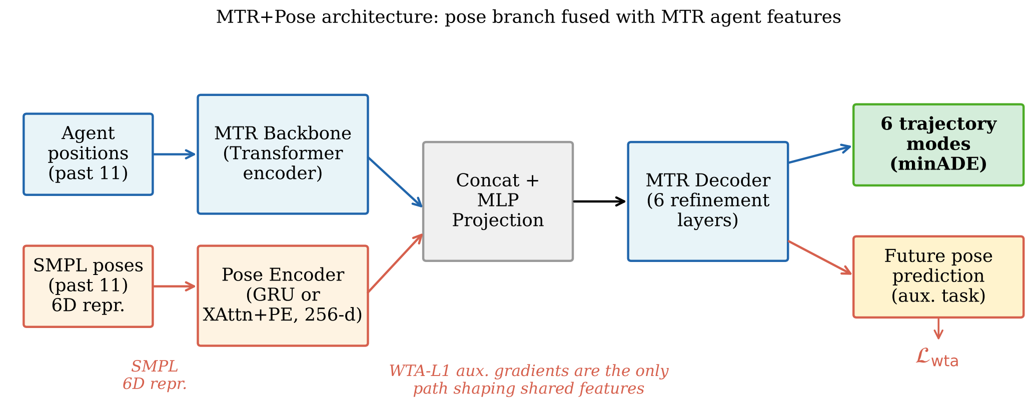

Figure 1: **MTR+Pose architecture.** A GRU pose encoder processes the past 11 SMPL body poses (each represented as a 144D 6D-rotation vector) and produces a 256D hidden state that is concatenated with the MTR agent feature and projected before the decoder. Auxiliary future-pose prediction heads are trained with a winner-takes-all L1 loss $\mathcal{L}_{\text{wta}}$ on 6D rotations; gradients from this auxiliary task are the only path by which pose supervision shapes the shared features. The trajectory decoder and map attention modules of MTR are unchanged.

- **Winner-takes-all L1 (WTA-L1)**: $\mathcal{L}_{\text{wta}}$ selects, per pedestrian, the pose mode closest to the ground truth in 6D L2 distance, and applies an L1 regression loss to that mode, masked by per-timestep trajectory validity. (In our configuration files this loss is named `gmm_pose` for historical reasons; it is a plain WTA regression, not a Gaussian mixture likelihood.)
- **Geodesic loss**: $\mathcal{L}_{\text{geo}} = \frac{1}{IJ}\sum_{i,j} \arccos\left(\text{clamp}\left(\frac{\text{tr}(\mathbf{R}_i^{j\top}\hat{\mathbf{R}}_i^j)-1}{2}, -1+\epsilon, 1-\epsilon\right)\right)$, where $\mathbf{R}, \hat{\mathbf{R}} \in SO(3)$ are rotation matrices recovered from the 6D representations of the WTA-selected mode, averaged over all future steps (unmasked). The clamp (ϵ=10^{-7}) avoids the divergent $\arccos$ derivative at the ± 1 boundary.
- **MPJPE**: $\mathcal{L}_{\text{mpjpe}} = \frac{1}{IJ}\sum_{i,j} \|\mathbf{x}_i^j - \hat{\mathbf{x}}_i^j\|_1$, where $\mathbf{x}$ are joint positions computed via forward kinematics on SMPL.
- **Pose mode classification**: $\mathcal{L}_{\text{cls\_pose}}$, a cross-entropy loss on per-mode pose scores against the WTA-selected mode.

The total loss adds the three MTR motion terms ($\mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{reg}} + 0.5\,\mathcal{L}_{\text{vel}}$) to the weighted pose terms $w_{\text{wta}}\mathcal{L}_{\text{wta}} + w_{\text{geo}}\mathcal{L}_{\text{geo}} + w_{\text{mpjpe}}\mathcal{L}_{\text{mpjpe}} + w_{\text{cls\_pose}}\mathcal{L}_{\text{cls\_pose}}$; we ablate $w_\star \in \{0, 0.05, 0.1, 0.2\}$ (weights of 1.0 diverge to NaN through the SMPL forward pass, so all stable runs use $w \leq 0.2$).

A caveat discovered post-hoc: future pose ground truth is absent in the main data condition. In our primary (10 fps) data condition, SMPL pose observations cover only the *past* window ($t \leq 1.0$ s); future pose targets are zero for 99.99% of timesteps (5 of 91,520 in the training split). As we show analytically and empirically in Section 4.3, this renders the geodesic term *inert* (identically zero gradient against all-zero targets, which map to the all-zero matrix under Gram–Schmidt), while WTA-L1, MPJPE, and $\mathcal{L}_{\text{cls\_pose}}$ stay active but pull predictions toward the degenerate zero targets. We report it prominently because it reverses a conclusion an earlier draft drew from single-run comparisons.

3.3 Cross-Attention Pose Encoders (Ablation)

To isolate the contribution of temporal ordering from sequential integration, we test two cross-attention pose encoder variants. Let $\mathbf{q} \in \mathbb{R}^{256}$ be the MTR agent feature (projected query) and $\mathbf{K}, \mathbf{V} \in \mathbb{R}^{11\times 256}$ be the 11 projected pose frames (keys and values). The cross-attention output is:

$$\mathbf{c} = \text{softmax}\left(\frac{\mathbf{q}\mathbf{K}^\top}{\sqrt{256}}\right)\mathbf{V}, \tag{2}$$

and $\mathbf{c}$ is concatenated with $\mathbf{q}$ and projected to 256D, replacing $\mathbf{h}_T$ in the fusion step.

Cross-attention without positional encoding. No temporal information is injected before attention: $\mathbf{K} = \{W_K \mathbf{p}_t\}_{t=1}^{11}$, $\mathbf{V} = \{W_V \mathbf{p}_t\}_{t=1}^{11}$. This treats pose frames as a permutation-invariant set and tests whether the raw content of each pose frame—absent any ordering—is sufficient to condition trajectory prediction.

Cross-attention with sinusoidal positional encoding. Standard sinusoidal positional encoding (Vaswani et al., 2017) is added to the projected keys and values before attention: $\mathbf{K}' = \{W_K \mathbf{p}_t + \mathbf{e}_t\}_{t=1}^{11}$, where $\mathbf{e}_t$ is the fixed sinusoidal embedding for frame index $t \in \{0, \dots, 10\}$. This injects temporal ordering at zero extra parameter cost and tests whether ordering alone is sufficient, without the causal accumulation structure of a GRU.

4 Experiments

4.1 Dataset and Setup

Dataset. We use a pedestrian-only subset of the Waymo Open Motion Dataset (Ettinger et al., 2021) paired with SMPL body pose pseudo-ground-truth from the Waymo-3DSkelMo pipeline (Zhu et al., 2025). The subset contains 579 training scenes (1,348 pedestrians, of which 1,144 have sufficient valid past observations to serve as prediction targets) and 5,171 validation scenes (8,282 pedestrians). Ground-truth future trajectories come from Waymo tracking data (91 timesteps at 10 Hz; 9.0 s total span — 1.0 s past plus an 8.0 s future horizon). 6,688 of the 8,282 validation pedestrians (80.8%) have valid future trajectory data and enter the evaluation.

Pose provenance. The SMPL poses are not motion-capture ground truth: they are estimated from Waymo LiDAR returns by the Waymo-3DSkelMo pipeline (Zhu et al., 2025), which fits per-frame SMPL bodies with LiDAR-HMR (Fan et al., 2023) and temporally refines them with motion priors trained on AMASS (Mahmood et al., 2019; Rempe et al., 2021; He et al., 2022). Two properties of this pipeline matter for interpreting our results. *First*, pose observations are available only for the past window in our main data condition, so no future information leaks into the model inputs; the evaluation is sound. *Second*, the pipeline includes a root-orientation correction step that snaps the SMPL root rotation to the (past-trajectory) motion direction whenever the pedestrian is moving and the raw LiDAR estimate deviates from that direction by more than $90°$. The root-orientation channel of the pose input is therefore partially derived from the position history—information the model already receives—while body configuration, gait, and orientation detail within $\pm 90°$ are genuinely perception-derived. Gains from pose conditioning should be interpreted with this partial overlap in mind.

Evaluation. We report **minADE** (minimum Average Displacement Error over 6 modes, in meters) and **minFDE** (minimum Final Displacement Error). Lower is better. The evaluation covers the full 8.0 s future horizon (80 steps at 10 Hz).

Training and statistical protocol. All models train for 30 epochs with batch size 10 on a single RTX 4090 (AdamW, LR 10^{-4}, step decay $\times 0.5$ at epochs 10/20/25, gradient-norm clip 1.0); a full run takes 10–15 minutes. Validation is run at epochs 1, 2, 4, every second epoch up to 20, and every epoch from 21 to 30. Following common practice, our per-run headline number is the *best* validation minADE over these evaluations; because best-of-n selection on a noisy curve inflates apparent differences, every key comparison is additionally run with **three to five random seeds** (five for the headline cells: baseline, WTA-L1/GRU, and the no-auxiliary GRU control), and we report mean $\pm$ std across seeds together with **paired bootstrap** 95% confidence intervals over the 6,688 evaluated pedestrians (per-pedestrian errors averaged across seeds, resampled jointly for both models in a pair; 10,000 resamples).

Scale caveat. The 579-scene training subset is a small fraction of the full Waymo training set ($\approx$487k scenarios), so absolute minADE values in Sections 4.2–4.5 are far from full-data state of the art; relative, seed-replicated comparisons between models trained on identical data are the object of study. Section 4.6 quantifies the scale effect directly.

Table 1: Main result (no map; best-checkpoint protocol; 5 seeds for the first three rows, 3 for the last). Per-seed values expose a heavy-tailed failure mode in GRU-based pose conditioning that mean±std alone understates.

Model	Aux. loss	minADE ↓ (mean±std)	per-seed values
MTR (no pose)	—	0.6604 ± 0.0143	.655 .683 .664 .654 .646
MTR+POSE GRU, no aux	none active	0.6545 ± 0.0271	.696 .665 .639 .626 .647
MTR+POSE GRU, WTA-L1	w=0.1	0.6853 ± 0.0651	.639 **.785** .629 **.717** .657
MTR+POSE XAttn+PE, WTA-L1	w=0.1	$\mathbf{0.6371 \pm 0.0026}$	.640 .634 .637

4.2 MAIN RESULT: POSE CONDITIONING VS. BASELINE

Table 1 compares, over three to five seeds each, the trajectory-only MTR baseline against three pose-conditioned variants: the GRU encoder with the WTA-L1 auxiliary pose loss (w=0.1; the configuration an earlier draft mislabeled "geodesic-only", see Section 4.3), the GRU encoder with *no* active auxiliary pose loss, and the cross-attention encoder with sinusoidal positional encoding (same WTA-L1 auxiliary loss).

The seed-replicated picture inverts the earlier draft's headline. The GRU encoder with the WTA-L1 auxiliary loss—credited with -7.6% from one lucky run—is *worse than the baseline on average*: two of five seeds converge to clearly degraded optima (bold), and the paired bootstrap over pedestrians puts it at $+3.8\%$ minADE (95% CI $[+3.2, +4.3]$). Its best seeds (0.629, 0.639) do beat every baseline seed, but no single-seed experiment can tell a lucky draw from a stable improvement. The cross-attention encoder with positional encoding delivers what the draft attributed to the GRU: a stable -3.5% minADE (CI $[-4.0, -3.0]$, $p{<}10^{-4}$; minFDE 1.399 ± 0.023 vs. baseline 1.539 ± 0.207), with per-seed spread $5.5\times$ smaller than the baseline's ($\sigma\,0.0026$ vs. 0.0143). The no-auxiliary GRU control edges out the baseline under the pedestrian bootstrap (-0.9%, CI $[-1.4, -0.4]$), but this margin does *not* survive a hierarchical bootstrap that also resamples seeds (CI $[-2.5, +1.9]\%$, p=0.59); we return to it in Section 4.3.

4.3 THE AUXILIARY-LOSS AUDIT: AN INERT TERM CREDITED WITH THE HEADLINE GAIN

An earlier draft ablated the pose supervision with one run per configuration and concluded geodesic-only supervision was the critical ingredient (0.6231, -7.6% vs. baseline 0.6745), beating "WTA-L1 only" (0.6467). Three facts reverse that conclusion.

Fact 1: the geodesic term received zero gradient. In the 10 fps data condition, future pose targets are zero for all but 5 of 91,520 training steps (Appendix A.2). A zero 6D target maps under Gram–Schmidt to the all-zero matrix $\mathbf{R}_{\text{gt}}{=}\mathbf{0}$, for which $\text{tr}(\hat{\mathbf{R}}^\top \mathbf{R}_{\text{gt}}) \equiv 0$ for *any* prediction, so $\mathcal{L}_{\text{geo}} \equiv \arccos(-\frac{1}{2})$, a constant; we verified its gradient is exactly zero (and nonzero under random targets).

Fact 2: the two compared configurations were therefore identical. The "geodesic-only" run also carried WTA-L1 at w=0.1, so with the geodesic term inert it ran the same training distribution as "WTA-L1 only"—their 0.0236 gap (0.6231 vs. 0.6467) is pure run-to-run noise, larger than most effects the draft reported.

Fact 3: seed replication measures the noise directly. Re-run with multiple seeds (Table 2), the pooled **eight** replicates of this single configuration span minADE 0.629–0.897 (0.7060 ± 0.0921); both draft values sit comfortably inside this one-configuration spread, confirming the "geodesic effect" was a draw from it.

The corrected interpretation reverses the draft on a second point as well: far from being "the critical ingredient", the auxiliary losses are—with the GRU encoder and degenerate zero-valued targets—associated with *instability*. Four of eleven runs across the WTA-L1 and MPJPE+WTA-L1 cells converged badly (≥ 0.71), whereas the no-auxiliary control produced no such seed in five runs and is the only GRU cell significantly better than the baseline (paired bootstrap -0.9%, CI $[-1.4, -0.4]$).

Table 2: Pose supervision ablation, seed-replicated (no map, GRU encoder, w=0.1 for active losses). "No aux" has no active auxiliary pose loss (the inert geodesic term only), so the pose encoder is trained by trajectory gradients alone. The WTA-L1 row pools the draft's "geodesic-only" and "WTA-L1 only" cells, which are functionally identical. Single-run draft values shown for comparison.

Aux. pose loss	minADE (mean±std)	per-seed values	single-run (draft)
None (no pose)	0.6604 ± 0.0143	.655 .683 .664 .654 .646	0.6745
No aux (inert geo)	0.6545 ± 0.0271	.696 .665 .639 .626 .647	—
WTA-L1 (8 seeds, pooled)	0.7060 ± 0.0921	.639 .785 .629 .717 .657 .897 .662 .662	0.6467 / 0.6231[†]
MPJPE + WTA-L1	0.6769 ± 0.0421	.724 .662 .644	0.6576
All (MPJPE+WTA+cls)	0.6574 ± 0.0237	.668 .674 .630	0.6532

[†]The earlier draft reported these as two different configurations ("WTA-L1 only" and "geodesic-only"); they are the same configuration, and their spread is seed noise.

Table 3: Temporal encoding ablation, seed-replicated (no map; WTA-L1 auxiliary loss at w=0.1 in all pose-conditioned variants).

Pose Encoder	Temporal info	minADE (mean±std)	per-seed values
None (no pose)	—	0.6604 ± 0.0143	.655 .683 .664 .654 .646
Cross-attention, no PE	none (bag)	0.6667 ± 0.0407	.714 .641 .646
Cross-attention + sinus. PE	ordering only	$\mathbf{0.6371 \pm 0.0026}$	.640 .634 .637
GRU (sequential)	ordering + causal	0.6853 ± 0.0651	.639 .785 .629 .717 .657

At our statistical power the active-loss variants cannot be usefully ranked against one another. The earlier draft's mechanistic story about geodesic gradients shaping the GRU was fitted to noise; the encoder comparison in Section 4.4 turns out to matter far more.

4.4 Ablation: Temporal Encoding — GRU vs. Cross-Attention

Table 3 compares three temporal encoding strategies (three to five seeds each), all with the WTA-L1 auxiliary loss at w=0.1.

Two conclusions survive seed replication, and one draft conclusion reverses. *Ordering is necessary*: the bag-of-frames encoder is slightly worse than the no-pose baseline ($+1.0\%$, CI $[+0.4, +1.6]$)—unordered pose content contributes nothing—while adding only a fixed sinusoidal positional encoding turns the same architecture into the best no-map model (-4.5% vs. no-PE, CI $[-5.2, -3.7]$; -3.5% vs. baseline). *The draft's ranking of GRU over attention reverses*: the GRU trails cross-attention+PE by $+7.6\%$ (CI $[+6.9, +8.2]$) once its failure seeds are accounted for, and even its best seeds only match it. The draft's "79% recovered by PE, a further 1.6% from sequential integration" decomposition was an artifact of comparing one lucky GRU run against one ordinary cross-attention run. The corrected statement: temporal ordering carries the pose signal, and the attention-based encoder extracts it more reliably than the GRU—with near-identical parameter counts ($\approx$497K vs. $\approx$505K).

4.5 Ablation: HD Map Context

The experiments above use zero-padded map features (the SMPL-paired pedestrian subset was originally processed without HD map polylines). To quantify how map context interacts with pose conditioning, we enable HD map loading and train the 2×2 factorial design: ±map × ±pose (GRU encoder, WTA-L1 auxiliary loss at w=0.1), three seeds per cell.

Table 4 reveals three findings. *First*, HD map context is the dominant signal: adding map features alone reduces minADE by 26.2% (CI $[-27.8, -24.6]$), far more than any pose effect (largest: 3.5% from xattn+PE), since pedestrian routing is encoded directly in the map. *Second*, with map context the incremental pose benefit shrinks to -0.45% (CI $[-0.90, -0.01]$, p=0.046 under the pedestrian bootstrap; not significant once seeds are resampled)—partial information redundancy, since body orientation and the map's routing constraints both encode heading, and part of the root-orientation

Table 4: Map × pose ablation, seed-replicated (three seeds per cell; WTA-L1 auxiliary loss at w=0.1 in pose cells). With map context the seed variance collapses by an order of magnitude relative to the no-map cells.

Pose Enc.	Map	minADE (mean±std)	per-seed values
None	No map	0.6604 ± 0.0143	.655 .683 .664 .654 .646
None	With map	0.4876 ± 0.0033	.490 .484 .489
GRU	No map	0.6853 ± 0.0651	.639 .785 .629 .717 .657
GRU	With map	$\mathbf{0.4854 \pm 0.0057}$	.480 .491 .485

Table 5: Pretrain × pose ablation, seed-replicated (three seeds per cell). All runs include HD map and use 30-epoch fine-tuning on the 579-scene subset (the no-pretrain rows repeat Table 4's map cells). Pretrained variants load the same full-Waymo backbone checkpoint via residual zero-init pose fusion; fine-tuning seeds vary.

Pretrain	Pose Enc.	minADE (mean±std)	minFDE (mean±std)	per-seed minADE
None	—	0.4876 ± 0.0033	1.0572 ± 0.0090	.490 .484 .489
None	GRU	0.4854 ± 0.0057	1.0362 ± 0.0060	.480 .491 .485
Full Waymo	—	0.3791 ± 0.0033	0.8019 ± 0.0025	.382 .375 .380
Full Waymo	GRU	$\mathbf{0.3718 \pm 0.0059}$	$\mathbf{0.7809 \pm 0.0097}$	.374 .376 .365

channel is derived from past heading to begin with (Section 4.1). *Third*, map context stabilizes training: per-seed spread in the map cells ($\sigma \leq 0.006$) is far smaller than in the no-map cells (σ up to 0.135; Table 7), and the pose-training failure mode of Section 4.2 did not occur in any of the six map-enabled seeds.

4.6 Ablation: Backbone Pretraining on Full Waymo

The 579-scene subset used above is ≈0.12% of the full Waymo training set (≈487k scenarios; 486,995 exactly). To test whether dataset scale is the binding constraint on absolute accuracy, and to isolate pose's contribution when the backbone already encodes a strong trajectory prior, we run a final 2×2 ablation: ±pretrain × ±pose, with HD map enabled in all four cells.

We pretrain the trajectory-only backbone (no pose encoder) on the full Waymo pedestrian set—231,507 per-agent training examples drawn from 486,995 scenarios—for 30 epochs with the same hyperparameters as before. To avoid corrupting this backbone at fine-tune iter 0, we use a *residual zero-init* pose fuser: its output is *added* to the backbone feature and its final linear layer is zero-initialized, so the pose pathway contributes exactly zero initially (the backbone passes through unchanged) and warms up additively.

Both pretraining conclusions survive seed replication cleanly. Pretraining reduces minADE by 22.3% without pose (CI $[-24.5, -20.1]$) and 23.4% with pose (CI $[-25.8, -21.1]$)—by far the largest effect after map context, and essentially independent of pose. On top of the pretrained backbone, pose conditioning contributes -1.9% minADE (pedestrian bootstrap CI $[-2.8, -1.0]$; two of three seeds improve, one flat, 0.3753 vs. 0.3760) and -2.6% minFDE—smaller than the draft's single-run 2.9%. This is pose's largest realistic (map-equipped, well-trained) gain, but only *marginal* once seeds are resampled (hierarchical bootstrap p=0.10); the draft's claim that pose's benefit *increases* with pretraining holds at the point estimates (-0.45% scratch vs. -1.9% pretrained) but neither is seed-level significant at three seeds (p=0.62, p=0.10)—suggestive, not established. Notably, the GRU encoder—unstable from scratch without a map—is well behaved in all six fine-tuning seeds, consistent with the residual zero-init fuser warming up the pose pathway against a strong trajectory prior. Qualitative examples at pretrained scale are in Appendix A.7, Figure 4.

5 Analysis and Insights

How an inert loss became a headline result. The failure chain is worth spelling out because none of its links is exotic: a data condition silently degenerated (all-zero future-pose targets), the loss evaluated to a smooth constant with *zero* gradient rather than crashing, two functionally identical configurations were each run once, and the resulting 0.024 seed gap was read as a 3.8% design effect and given a mechanistic story. The audit that caught it generalizes to two cheap checks: verify the *target coverage* of every auxiliary loss against the data, and check the *gradient* of every loss term in isolation on a real batch.

What makes the pose branch work. The seed-replicated evidence points at the encoder's temporal structure, not the auxiliary supervision. Unordered pose content is useless (bag-of-frames $\approx$ baseline), so the exploited signal is *temporal*—gait phase and orientation change, not static posture, and (Section 4.4) attention extracts it more reliably than the GRU. The mechanism is pose *encoding*, not *prediction*: real dense future-pose targets—the only condition in which the rotation-space losses carry gradient—still do not move accuracy (Appendix A.5), and the auxiliary WTA-L1 loss is not required for a gain (the no-auxiliary control is -0.9%). Its degenerate zero targets are plausibly counterproductive, anchoring shared features toward a constant, which may explain why the 20% heavy-tailed failure rate of no-map pose runs (bad basins, not late divergence) concentrates in auxiliary-active GRU cells and vanishes with map context or pretraining—reinforcing that on small benchmarks per-seed values or failure rates must be reported, not seed means alone.

6 Limitations

Dataset scale and transfer. Our main ablations (Sections 4.2–4.5) use a 579-scene subset; Section 4.6 addresses this only partially, since fine-tuning still uses that subset. The point estimates suggest a pose benefit that grows with backbone quality (-0.45% from scratch vs. -1.9% pretrained), but neither is seed-level significant at three seeds (p=0.62, p=0.10); a definitive answer requires training the pose-conditioned model on full Waymo, beyond our compute budget.

Pose provenance and decoder. The LiDAR-derived SMPL pseudo-ground-truth (Waymo-3DSkelMo) is matched to Waymo tracks; its root-orientation correction partially couples the root channel to past trajectory (Section 4.1), so part of the pose signal overlaps with observed position history, and online pose estimation in deployment would carry different noise. We also do not evaluate predicted-pose quality independently—our metric is trajectory only.

Statistical power and two bootstrap levels. Our primary paired bootstrap resamples pedestrians on *seed-averaged* errors; we also report a *hierarchical* bootstrap resampling seeds then pedestrians—the conservative test we rely on for significance. Under it the encoder (-3.5%), map (-26%), and pretraining (-22%) effects survive ($p{\leq}0.01$), but the small pose margins do not (p=0.62, p=0.10; three seeds). Three seeds also cannot rank the active auxiliary losses, and the best-checkpoint protocol (kept for comparability) selects on validation noise. All CIs cover one validation split only.

7 Conclusion

We have presented MTR+Pose, a pose-conditioned extension of the Motion Transformer that incorporates past SMPL body poses via a GRU encoder with 6D rotation representations, together with a seed-replicated measurement of what pose conditioning is worth and an audit that retracts our own earlier headline claim. Our conclusions: (1) the previously credited geodesic SO(3) supervision was inert—zero gradient against absent future-pose targets—so its 7.6% was seed noise between two identical configurations. (2) Whether pose helps is decided by the *encoder*: cross-attention with sinusoidal positional encoding gives a stable -3.5% minADE (CI $[-4.0, -3.0]$), while the GRU is heavy-tailed and worse than baseline on average, and temporal ordering—not the auxiliary loss or sequential integration—carries the signal. (3) The mechanism is the past pose encoder, not future-pose prediction. (4) Context dominates: HD map alone is worth -26% and full-Waymo pretraining a further $\sim$22% (both seed-robust), while the pose gains they leave (-0.45%, -1.9%) are not seed-level significant at three seeds—the audit methodology behind these corrections, not the pose numbers, is our intended lasting contribution.

Reproducibility Statement

All experiments use the publicly available Waymo Open Motion Dataset and the open-source Waymo-3DSkelMo pose-extraction pipeline; neither is redistributed by us. Training configurations (YAML), the multi-seed launch script, the per-pedestrian metric logger, and the bootstrap analysis script are included in our code release, together with the exact random seeds (101/202/303, plus 404/505 for the headline cells: baseline, WTA-L1/GRU, and the no-auxiliary GRU control) and the per-epoch validation protocol described in Section 4.1. Appendix A lists all hyperparameters; Appendix A.2 documents dataset construction, including the timestamp-alignment and future-pose-coverage details on which the inert-gradient analysis rests. We will release the code upon publication.

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LLM Disclosure

The authors used large language model (LLM) assistance (Claude, Anthropic) during the preparation of this manuscript for writing drafts, code assistance, and literature search support. All experimental results, scientific analysis, and conclusions are the authors' own.

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A ADDITIONAL EXPERIMENTAL DETAILS

A.1 ARCHITECTURE HYPERPARAMETERS

- GRU pose encoder: GRU(input=144, hidden=256, layers=2, dropout=0.1); ≈505K parameters total including fusion MLP
- Cross-attention pose encoder: pose projection 144→256, MultiheadAttention(embed_dim=256, num_heads=8, dropout=0.1, batch_first=True); ≈497K parameters total including fusion MLP
- Pose input dimension: 144 (24 joints × 6D rotation)
- Past pose window: 11 timesteps (1.0 second at 10 Hz)
- Number of predicted trajectory modes: $K = 6$
- Number of future steps: 80 (8.0 seconds)
- MTR context encoder: 6 local-attention transformer layers, 256 hidden dim, 8 heads, attention dropout 0.1
- MTR decoder: 6 layers, hidden dim 512 (map branch 256), 8 heads
- Batch size: 10
- Optimizer: AdamW, $\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 0.01
- LR schedule: initial LR 1×10^{-4}, step decay $\times 0.5$ at epochs 10, 20, 25
- Epochs: 30
- Gradient clipping: norm ≤ 1.0
- Random seeds for replicated runs: 101, 202, 303 (plus 404, 505 for headline cells)

A.2 DATASET CONSTRUCTION

Waymo agent future trajectories are stored in `processed_scenarios/` PKL files with shape `(num_agents, 91, 10)`. SMPL pedestrian files are matched to Waymo object IDs via integer stem matching. 6,688 of 8,282 validation pedestrians (80.8%) have a valid future-trajectory match and are evaluated; the remainder are excluded. The 91-step grid spans $[0, 9.0]$ seconds; SMPL timestamps (originally in microseconds) are divided by 10^6 before alignment. In the 10 fps condition the SMPL observations cover only $[0, 1.0]$ s (the past window): across the entire training split only 5 of 91,520 future pose-target steps are nonzero, which is the basis for the inert-geodesic-gradient analysis in Section 4.3.

A.3 LOSS WEIGHT STABILITY

At $w = 1.0$, gradient explosion through the SMPL forward kinematics causes NaN divergence at epoch 3. The geodesic loss gradient norm was also found to diverge at the ± 1 boundary of the $\arccos$ argument; we clamp to $[-1 + \epsilon, 1 - \epsilon]$ with $\epsilon = 10^{-7}$. At $w = 0.1$, training is stable for all 30 epochs across all ablation variants.

A.4 AUXILIARY LOSS WEIGHT

The earlier draft swept what it believed was the geodesic weight and found a "sharp optimum" at w=0.1: w=0.05 gave 0.6786 and w=0.2 gave 0.6660, both single runs. Two corrections apply. First, because the geodesic term is inert, the sweep actually varied the *WTA-L1* weight (both weights were scaled together in those configurations). Second, the differences involved (0.02–0.06) overlap the

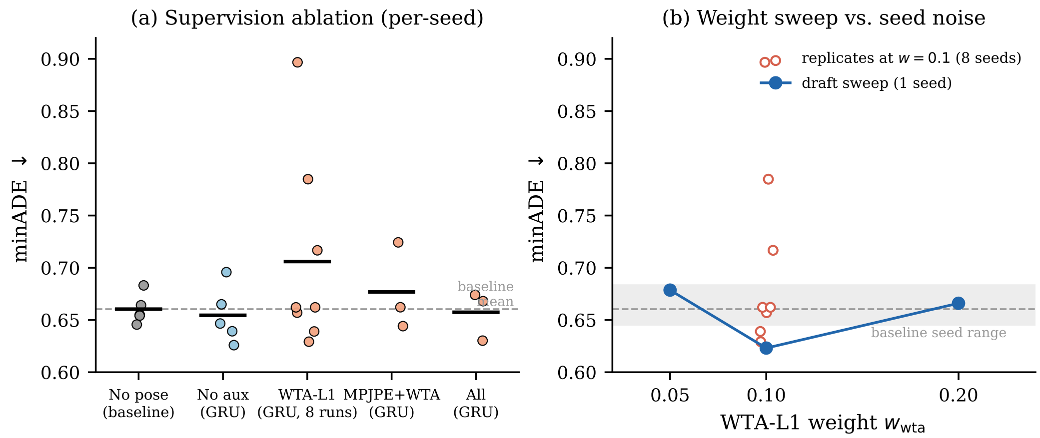

Figure 2: **Loss ablation and WTA-L1 weight sensitivity.** *Left*: minADE for the seed-replicated supervision configurations (Table 2); markers show individual seeds. *Right*: single-seed sensitivity to the auxiliary WTA-L1 weight ($w \in \{0.05, 0.1, 0.2\}$), with the seed-noise scale measured in Section 4.3 shown as a shaded band. Within that noise, the apparent optimum at w=0.1 is weak evidence at best.

seed-noise scale measured in Section 4.3, so the curvature of this one-seed sweep should not be read as a calibrated sensitivity estimate. What survives: w=0.1 is a reasonable default, and we did not find evidence that performance is improved by halving or doubling it.

A.5 Ablation: Future Pose Supervision Quality

To localize the source of improvement, we compare two data conditions, both with the full auxiliary loss set at w=0.1 (single runs): (a) *no future poses*: future pose GT is zero, only past poses are observed (the same data condition as our other ablations); and (b) *30 fps motion-prior-completed future poses*: the Waymo-3DSkelMo pipeline's motion-prior infilling provides 99% coverage of future timesteps as pseudo-GT. Condition (b) is the one setting in which the geodesic and MPJPE terms receive real (non-degenerate) targets and gradients.

The result: the 30 fps model achieves minADE $= 0.6567$ and the zero-future model 0.6532—a difference of 0.0035 (0.5%), well within the seed noise measured in Section 4.3. **Supervising the pose decoder with real future poses does not improve trajectory prediction**: even when the rotation-space losses are given genuine gradients, trajectory accuracy does not move. Whatever benefit pose conditioning provides must come from the past GRU encoder, not from the auxiliary future-pose prediction task.

A.6 Training Dynamics

Figure 3 shows validation minADE over 30 training epochs for the temporal encoding variants, with per-seed bands. Two qualitative observations are stable across seeds: cross-attention without positional encoding tracks the trajectory-only baseline throughout, and the pose-conditioned variants exhibit larger epoch-to-epoch variability than the baseline. The GRU band is dominated by its failed seeds, whose validation curves plateau near 0.72–0.83 from early in training rather than diverging late—the failure is a bad basin, not a late-training collapse—while the cross-attention+PE band is the narrowest of all variants.

A.7 Qualitative Examples

A.8 Original Single-Run Results (Earlier Draft)

Table 6 reports all 19 single-seed runs of the earlier draft, with loss labels corrected per the audit: "geo" terms were inert (zero gradient, Section 4.3), so the functional supervision is what remains after

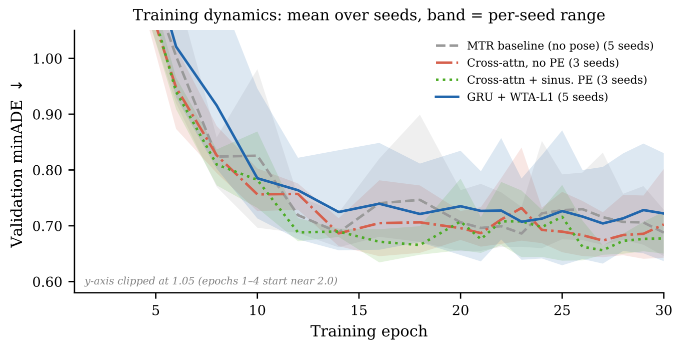

Figure 3: **Training dynamics.** Validation minADE over training for the temporal encoding variants; lines are means over seeds, bands the per-seed range. Epoch-to-epoch and seed-to-seed variability is of the same order as the differences between several variants, which is why all comparisons in this paper are seed-replicated.

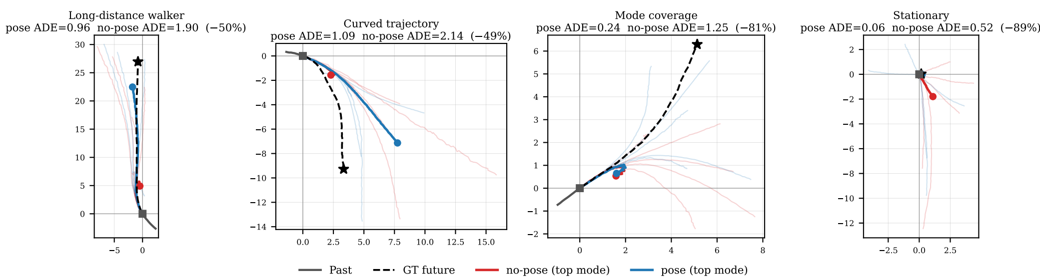

Figure 4: **Qualitative comparison at full-Waymo-pretrained scale.** Four validation scenes. Past in gray; GT future dashed black with ⋆ endpoint. All 6 predicted modes faded; top-scoring mode highlighted (red: no-pose; blue: pose). Per-scene minADE in titles. Scenes selected from the largest per-pedestrian improvements of the pose model, so these illustrate the failure modes pose can fix, not typical magnitudes. *Long-distance:* no-pose stops at 5m; pose follows the 25m walk. *Curved:* no-pose underestimates displacement; pose follows the curve. *Mode coverage:* both top modes underestimate, but pose has a faded mode aligned with GT. *Stationary:* no-pose wrongly predicts motion; pose correctly stays near origin.

removing them. Runs 009 and 010 are functionally identical; their spread is seed noise. These runs use unrecorded seeds and are superseded by the seed-replicated matrix of Table 7.

A.9 SEED-REPLICATED MATRIX

Table 7 lists every run of the replication matrix (seeds 101/202/303, plus 404/505 for the headline cells).

Table 6: Earlier-draft single runs (corrected labels). "Functional losses" removes inert terms; w in parentheses. Baseline for Δ is run 004. Runs 001–002 are invalid (zero future trajectory GT).

Run	Draft tag	Encoder	Functional losses / Map / Pretrain	minADE	Δ
001	full pose	GRU	all (1.0)	0.6114	*invalid*
002	no pose	—	traj only	0.7724	*invalid*
003	full	GRU	mpjpe+wta+cls (0.1)	0.6532	-3.2%
004	baseline	—	traj only	0.6745	—
005	full w=0.05	GRU	mpjpe+wta+cls (0.05)	0.6557	-2.8%
006	full w=0.2	GRU	mpjpe+wta+cls (0.2)	0.6542	-3.0%
007	30fps	GRU	mpjpe+geo+wta+cls (0.1), real future	0.6567	-2.6%
008	"mpjpe only"	GRU	mpjpe+wta (0.1)	0.6576	-2.5%
009	"wta only"	GRU	wta (0.1)	0.6467	-4.1%
010	"geo only"	GRU	wta (0.1) [geo inert]	0.6231	-7.6%
011	"geo w=0.05"	GRU	wta (0.05) [geo inert]	0.6786	$+0.6\%$
012	"geo w=0.2"	GRU	wta (0.2) [geo inert]	0.6660	-1.3%
013	cross-attn	XAttn	wta (0.1) [geo inert]	0.6765	$+0.3\%$
014	xattn+PE	XAttn+PE	wta (0.1) [geo inert]	0.6337	-6.0%
015	no-pose+map	—	traj only + map	0.4880	-27.7%
016	pose+map	GRU	wta (0.1) + map [geo inert]	0.4797	-28.9%
017	pretrain	—	traj only + map (full Waymo, 231k peds)	—	*ckpt only*
018	finetune pose	GRU	wta (0.1) + map, pretrained	0.3749	-44.4%
019	finetune no-pose	—	traj + map, pretrained	0.3861	-42.8%

Table 7: Seed-replicated matrix: best-checkpoint minADE per seed (seeds 101/202/303, plus 404/505 for headline cells). All runs 30 epochs, identical protocol; "geo (inert)" marks the zero-gradient geodesic term carried by some configurations.

Cell	Configuration	minADE mean±std	per-seed values
baseline	no pose	0.6604 ± 0.0143	.6550 .6831 .6641 .6541 .6456
geo_pure	GRU, no active aux (geo inert)	0.6545 ± 0.0271	.6958 .6650 .6392 .6259 .6466
wta01	GRU, WTA-L1 + geo (inert)	0.6853 ± 0.0651	.6390 .7848 .6292 .7167 .6570
gmm_only	GRU, WTA-L1 (same config as wta01)	0.7403 ± 0.1354	.8967 .6621 .6621
mpjpe	GRU, MPJPE + WTA-L1	0.6769 ± 0.0421	.7243 .6622 .6441
full	GRU, MPJPE+WTA+cls (geo inert)	0.6574 ± 0.0237	.6680 .6740 .6302
xattn	XAttn no PE, WTA-L1 (+ geo inert)	0.6667 ± 0.0407	.7136 .6408 .6457
xattn_pe	XAttn+PE, WTA-L1 (geo inert)	0.6371 ± 0.0026	.6395 .6343 .6374
map_nopose	no pose, with map	0.4876 ± 0.0033	.4903 .4839 .4886
map_wta01	GRU, WTA-L1, with map	0.4854 ± 0.0057	.4798 .4911 .4853
ft_nopose	no pose, map, pretrained	0.3791 ± 0.0033	.3815 .3753 .3804
ft_wta01	GRU, WTA-L1, map, pretrained	0.3718 ± 0.0059	.3744 .3760 .3651